

Sequences, Series, and the Binomial Theorem

Algebra 2

Evaluating binomial coefficients, expanding a binomial power in full, extracting one specific term, and using the sequence structure of a row of coefficients

Directions. The binomial theorem says

$$(u + v)^n = \sum_{k=0}^n \binom{n}{k} u^{n-k} v^k, \quad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Decide for each problem whether you need one coefficient, one term, or the whole expansion — the method is not the same.

Reference (rows of Pascal's triangle — no problem values here; take each problem's own numbers from its statement):

row $n = 3$:	1	3	3	1			
row $n = 4$:	1	4	6	4	1		
row $n = 5$:	1	5	10	10	5	1	
row $n = 6$:	1	6	15	20	15	6	1

1. Evaluate the binomial coefficient $\binom{6}{2}$ using the factorial formula. Show the factorials before you cancel.

Answer: _____

2. Evaluate $\binom{8}{3}$. Then state in one sentence what this number counts in the expansion of $(u + v)^8$.

Answer: _____

3. Expand $(x+2)^3$ completely, writing the coefficient of each term as a binomial coefficient times a power of 2 before you simplify.

Answer: _____

4. Row 5 of Pascal's triangle holds the coefficients of $(u+v)^5$. Add all six of those coefficients: $\binom{5}{0} + \binom{5}{1} + \cdots + \binom{5}{5}$. What is the total?

Answer: _____

5. In the expansion of $(x+2)^5$, find the **coefficient of** x^3 without expanding the whole thing. (Use the term $\binom{5}{k}x^k2^{5-k}$.)

Answer: _____

6. Expand $(2x - 1)^4$ completely. Watch two things: the 2 inside $2x$ gets raised to a power too, and the sign alternates.

Answer: _____

7. In the expansion of $(3 + y)^6$, find the **coefficient of y^2** . Note that here the constant 3 is the *first* term of the binomial.

Answer: _____

8. Find the sum of all the coefficients in the expansion of $(2x + 1)^5$. (Hint: adding every coefficient is the same as evaluating the expansion at a particular value of x — but write the sum $\sum_{n=0}^5 \binom{5}{n} 2^n$ and confirm it.)

Answer: _____

9. In the expansion of $(1+x)^n$ the coefficient of x^2 is $\binom{n}{2}$. For which n does that coefficient equal 45? Solve the resulting quadratic equation and say which root makes sense here.

Answer: _____

10. In the expansion of $(2x+y)^5$, find the **coefficient of** x^3y^2 . Both the binomial coefficient and the power of 2 matter.

Answer: _____

11. Expand $(a+3b)^3$ completely, in terms of a and b . Then say which entry of Pascal's triangle produced each coefficient.

Answer: _____

12. Synthesis. Consider $(3x - 2)^4$.

- (i) Write the general term of the expansion using $\binom{4}{n}$, a power of 3 and a power of -2 .
(ii) Add all the coefficients by evaluating that sum for $n = 0$ to $n = 4$. (iii) Explain in one sentence why the answer had to come out this way, using the value of x that makes every x -power equal 1.

Answer: _____